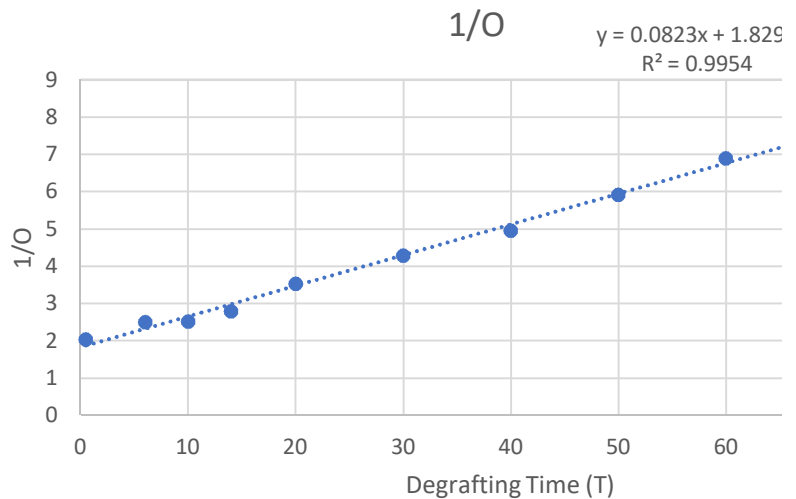


Table 1		
Degrafting time (min)	O (Chains/nm ²)	1/O
0.5	0.493	2.028398
6	0.401	2.493766
10	0.399	2.506266
14	0.358	2.793296
20	0.283	3.533569
30	0.234	4.273504
40	0.202	4.950495
50	0.169	5.91716
60	0.145	6.896552
70	0.131	7.633588



plot 1/O vs t, slope is k and intercept is 1/O	Linest Function		Solver	
	0.082291	1.829822	k	0.082291
	0.001981	0.074734	1/O	1.829822
From plot:	0.995385	0.142881	O	0.546501
k=0.0823	1725.599	8		
1/O = 1.8298	35.22827	0.163321		

The physical meaning of O_{not} is the initial density of chains, at $t=0$, it is 1.82.

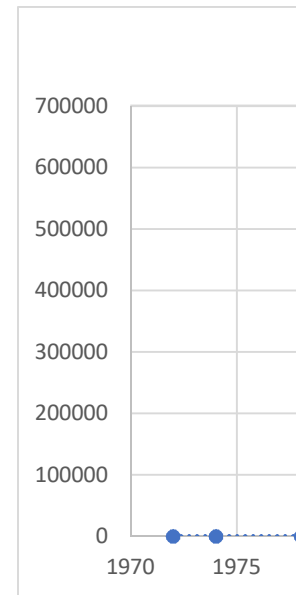
K is in nm²/min*chain, k is probably the inverse density of the reaction? If you increase the rate of reaction, k should decrease because it's in the denominator.

My data for part c is the same for part a. I believe that using a/b is a better method due to the lack of steps, meaning there is less chance I make a mistake in inputting.



For solver"	
Caluclated y's	Offset
1.870967349	0.024784
2.323566517	0.028968
2.652729548	0.021452
2.981892579	0.035569
3.475637125	0.003356
4.298544703	0.000627
5.12145228	0.029226
5.944359858	0.00074
6.767267436	0.016714
7.590175013	0.001885
Sum	0.163321

Processor	Year (Y)	Trasnsistors (T)	Using $T(Y+2)/T(Y)$
8008	1972	2.5	2
8080	1974	5	2.9
8086	1978	29	2.068965517
80286	1982	120	1.527777778
80386	1985	275	2.145454545
80486	1989	1180	1.313559322
Pentium	1993	3100	1.209677419
Pentium II	1997	7500	3.2
Pentium III	1999	24000	3.5
Itanium	2000	42000	3.492063492
Itanium 2	2003	220000	5.381818182
Itanium 2	2004	592000	
		Average	2.612665114



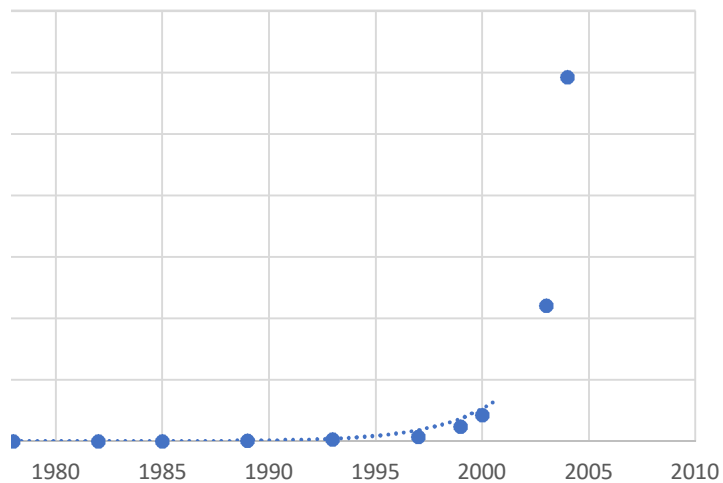
$\ln(T)$	Calcuated	
0.9162907	b=	0.354758699
1.6094379	$\ln(a)=$	-698.630203
3.3672958	a=	3.8793E-304
4.7874917		
5.6167711		
7.0732697		
8.0391574		
8.9226583		
10.085809		
10.645425		
12.301383		
13.291262		

From Trendline	
if $T = a \cdot e^{by}$	
a=	4.00E-304
b=	0.3548

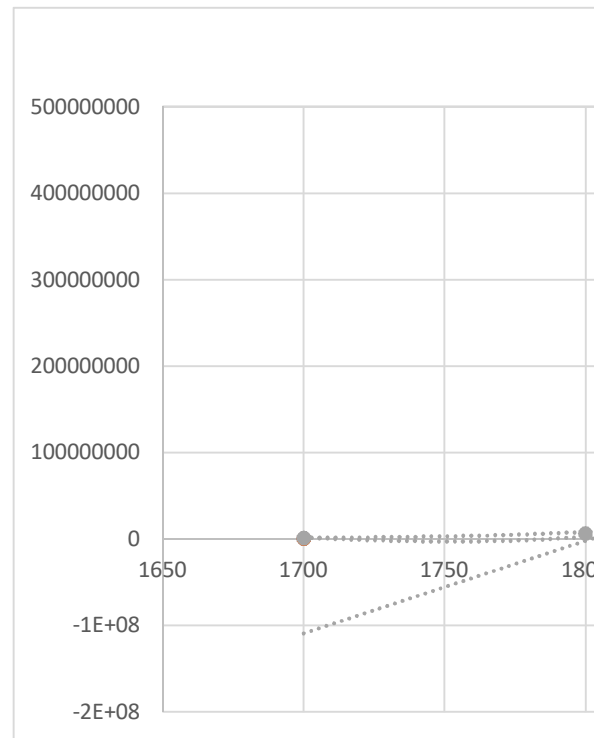
Note: for calculating $T(y+2)/T(y)$, I then divided by the difference in years/2, ensuring that the number was the ratio of change for two years over the gap.

Transistors

$$y = 4E-304e^{0.3548x}$$
$$R^2 = 0.9863$$



Year	USA (pop)
1500	2000000
1600	1500000
1650	
1750	
1700	1000000
1800	6000000
1810	7240000
1820	9980510
1830	13240314
1840	17443768
1850	23579718
1860	31838901
1870	40240630
1880	50458000
1890	63302000
1900	76391000
1910	92767000
1920	106881000
1930	123668000
1940	132637000
1950	152271000
1960	180671000
1970	205052000
1980	227726463
1990	250131894
2000	282158000

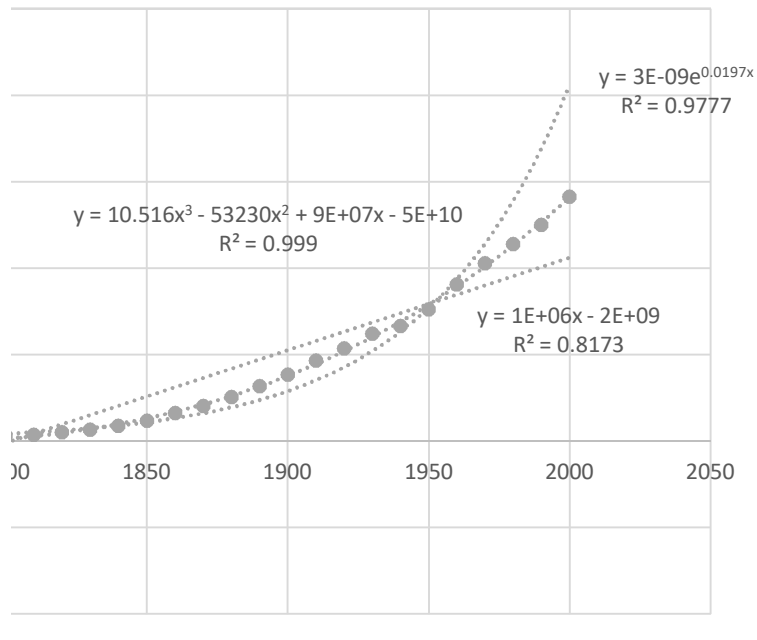


	1) Linear	2) Exponential
Slope	a= 1070926	a= 3E-09
Intercept	b= -1.9E+09	ln(b)= 0.0197
		b= 1.019895

	Projections for 300M		
Using	Linear	Exponential	Polynomial
Year	2082.134	1987.002	2006.113

The values predicted by the linear and exponential functions are way off. The linear function projects far into the future (60 years too late) and exponential says we already reached 300M (in 1987). The polynomial function fitted to a degree of three tells a much better story, the estimated prediction of 300M in 2006, which is only 4 years off the actual value of 328.3M in 2015, which is only 4 years off the actual value of 328.3 million in 2019.

Chart Title



3) Polynomial

ax^3	bx^2	cx	b
10.51579	-53230.3	89678384	-50280314163

off. Linear
the US
power of
106, and a
e of 328.3